

Math 1314 Lesson 16

1) Graph $y = x^2$, $y = x^2 + 2$, $y = x^2 - 3$

Complete the table of values.

x	$y = x^2$
-2	
-1	
0	
1	
2	

Graph 1

x	$y = x^2 + 2$
-2	
-1	
0	
1	
2	

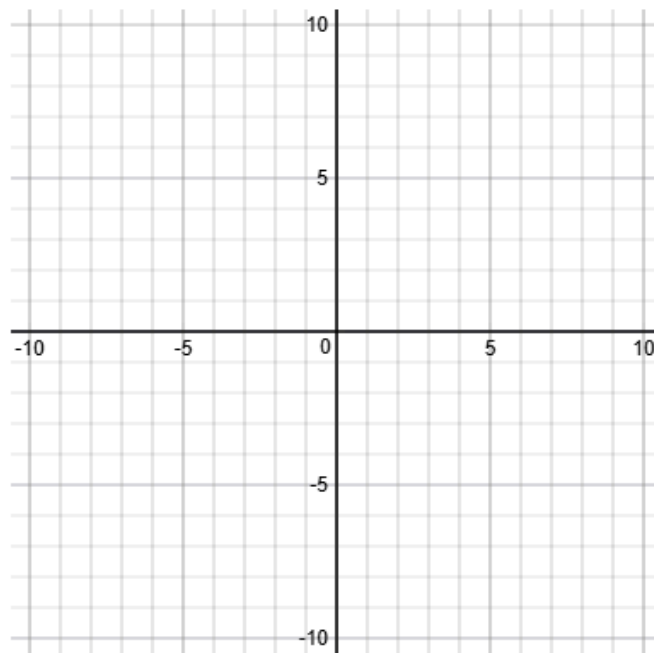
Graph 2

x	$y = x^2 - 3$
-2	
-1	
0	
1	
2	

Graph 3

How are Graph 1 and Graph 2 related?

How are Graph 1 and Graph 3 related?



2) Graph $y = x^2$ and $y = (x - 1)^2$, $y = (x + 3)^2$

Complete the table of values.

x	$y = x^2$
-2	
-1	
0	
1	
2	

Graph 1

x	$y = (x - 1)^2$
-1	
0	
1	
2	
3	

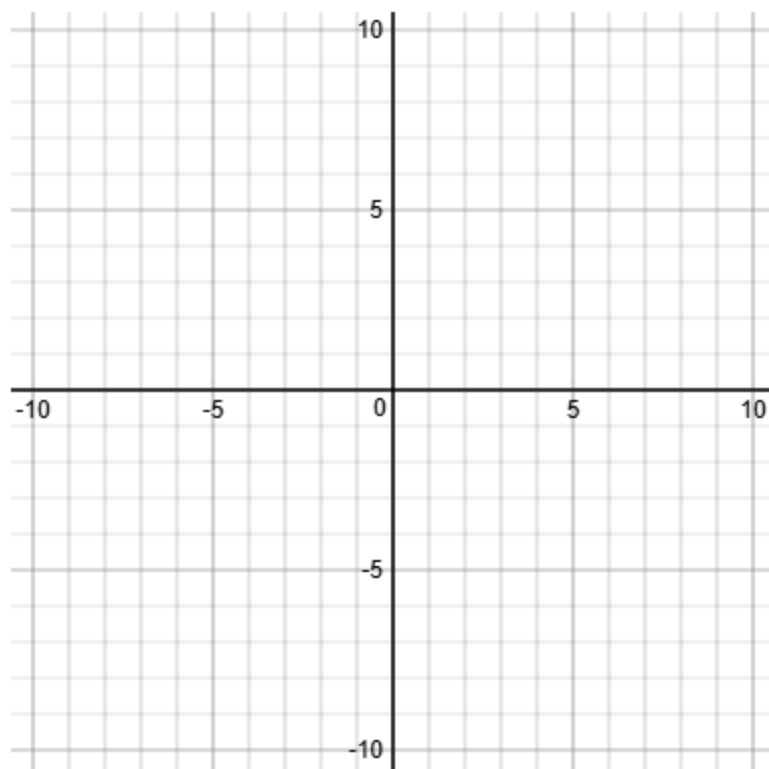
Graph 2

x	$y = (x + 3)^2$
-5	
-4	
-3	
-2	
-1	

Graph 3

How are Graph 1 and Graph 2 related?

How are Graph 1 and Graph 3 related?



3) Graph $y = x^3$, $y = x^3 + 2$, $y = x^3 - 3$

Complete the table of values.

x	$y = x^3$
-2	
-1	
0	
1	
2	

Graph 1

x	$y = x^3 + 2$
-2	
-1	
0	
1	
2	

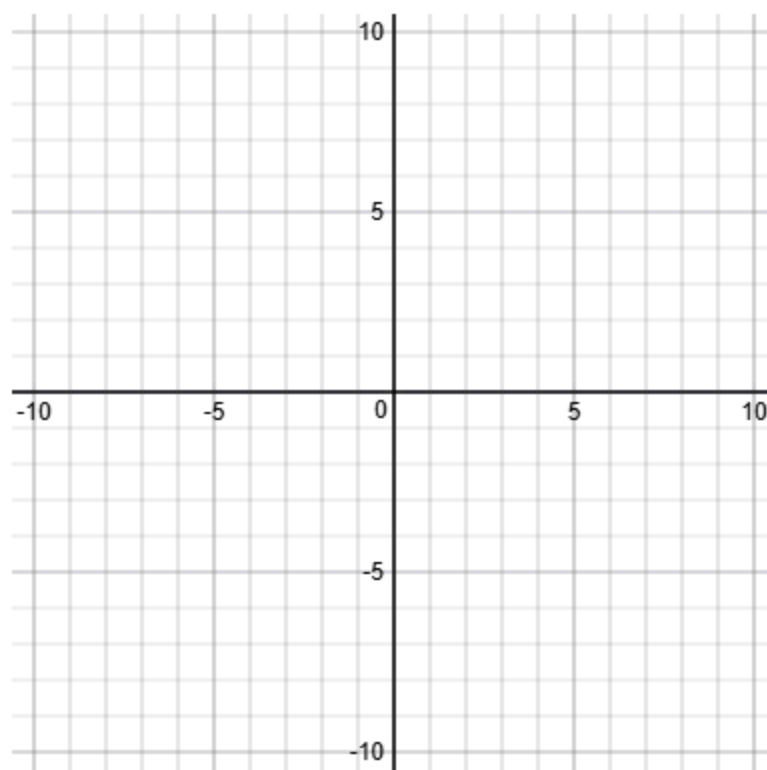
Graph 2

x	$y = x^3 - 3$
-2	
-1	
0	
1	
2	

Graph 3

How are Graph 1 and Graph 2 related?

How are Graph 1 and Graph 3 related?



4) Graph $y = x^3$ and $y = (x - 1)^3$, $y = (x + 3)^3$

Complete the table of values.

x	$y = x^3$
-2	
-1	
0	
1	
2	

Graph 1

x	$y = (x - 1)^3$
-1	
0	
1	
2	
3	

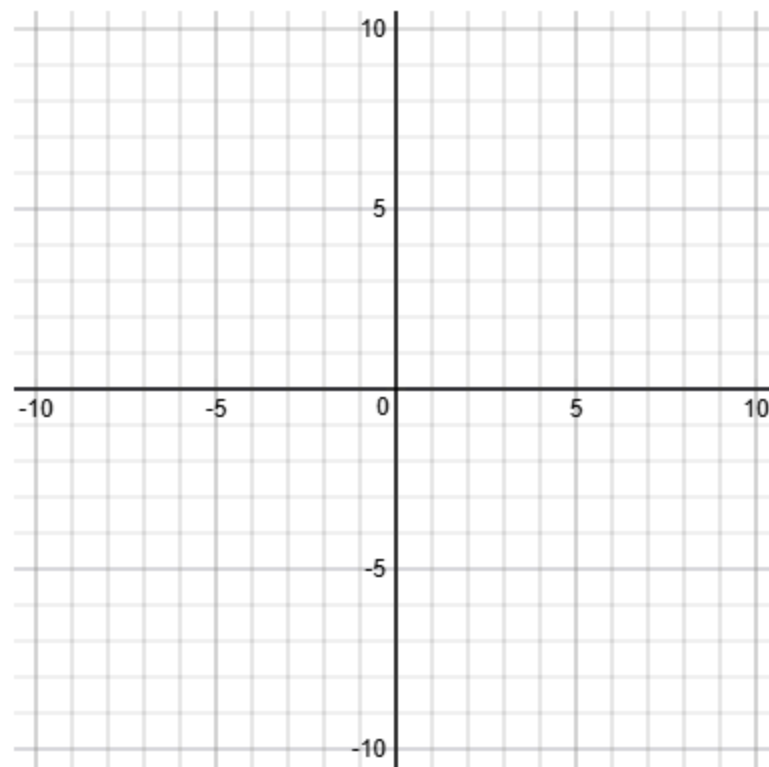
Graph 2

x	$y = (x + 3)^3$
-5	
-4	
-3	
-2	
-1	

Graph 3

How are Graph 1 and Graph 2 related?

How are Graph 1 and Graph 3 related?



5) Graph $y = \sqrt{x}$, $y = \sqrt{x} + 2$, $y = \sqrt{x} - 3$

Complete the table of values.

x	$y = \sqrt{x}$
0	
4	
9	
16	

Graph 1

x	$y = \sqrt{x} + 2$
0	
4	
9	
16	

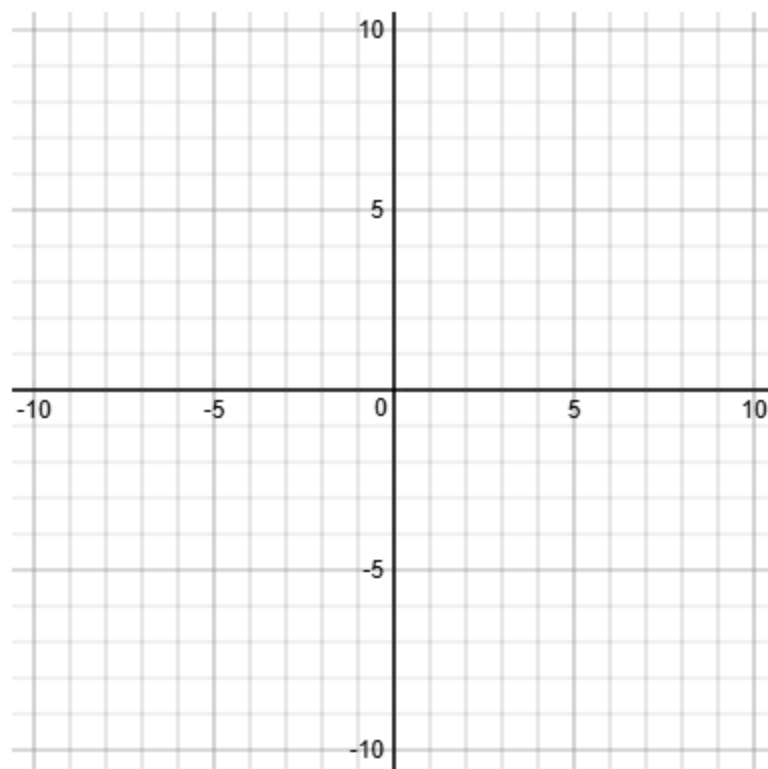
Graph 2

x	$y = \sqrt{x} - 3$
0	
4	
9	
16	

Graph 3

How are Graph 1 and Graph 2 related?

How are Graph 1 and Graph 3 related?



6) Graph $y = \sqrt{x}$ and $y = \sqrt{x-1}$, $y = \sqrt{x+3}$

Complete the table of values.

x	$y = \sqrt{x}$
0	
4	
9	
16	

Graph 1

x	$y = \sqrt{x-1}$
1	
5	
10	
17	

Graph 2

x	$y = \sqrt{x+3}$
-3	
1	
6	
13	

Graph 3

How are Graph 1 and Graph 2 related?

How are Graph 1 and Graph 3 related?

